Waste Management 104 (2020) 104–118

Contents lists available at ScienceDirect

Waste Management

journal homepage: www.elsevier.com/locate/wasman

The toxicity and valorization options of cigarette butts

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ARTICLE INFO

Article history:
Received 2 November 2019
Revised 9 January 2020
Accepted 10 January 2020
Available online 21 January 2020

Keywords:
Cigarette butts
Recycling
Environmental management
Cellulose acetate
Waste materials

ABSTRACT

Cigarette butts, one of the most abundant forms of waste in the world, contain more than 4000 toxic chemicals and pose serious risks to the health of wildlife, humans, and marine and freshwater organisms. Although trivial in size, trillions of cigarettes are produced every year worldwide, resulting in the accumulation of tonnes of toxic waste litter. In 2016, a world production of over 5.7 trillion cigarettes was reported with the majority comprising cellulose acetate filters – a polymer with poor biodegradability. Depending on the environmental conditions, cellulose acetate filters can take up to 10 years to decompose during which time they leach heavy metals and toxic chemicals into the environment. Although possible disposal methods for collected cigarette butt waste include incineration and landfilling, both techniques may result in the release of hazardous fumes and can be costly. However, recycling CBs in different materials could be a possible solution for this concurrent environmental pollution. A number of novel studies have been publicized on recycling cigarette butts with encouraging results, and several methods have been studied, including recycling of cigarette butts in asphalt concrete and fired clay bricks, as a carbon source, sound absorbing material, corrosion inhibitor, biofilm carrier, and many more. Hence, this paper provides a comprehensive review and discussion of various studies that have been carried out on the toxicity and valorization of cigarette butt waste and investigates the feasibility and sustainability of recycling methods adopted. Further research and developments are essential for the widespread application of recycling cigarette butts.

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https://doi.org/10.1016/j.wasman.2020.01.011
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1. Introduction

Cigarette butts (CBs) are the world's most prevalent source of waste litter. The Tobacco Atlas (2018) reported that during the year 2016, 5.7 trillion cigarettes were consumed worldwide and that about 97% of the cigarette filters were composed of cellulose acetate, a modified natural polymer (Puls et al., 2011). In Australia, it is estimated that over 20 billion filtered cigarettes are consumed each year, of which an estimated 7 billion are littered into the environment (Wallbank et al., 2017). That is 1 out of every 3 smoker litter their CBs into the environment. China is the leading consumer of cigarettes, smoking 41.9% of all cigarettes globally (The Tobacco Atlas, 2018). It is estimated that during the period 2015 to 2025, tobacco smokers are set to increase by 7 million in Nigeria and 24 million in Indonesia (The Tobacco Atlas, 2018).

To address the tobacco menace, certain developed countries have implemented strong tobacco control regulations and have seen significant reductions in smoking (Mackay et al., 2000). However, the decline in smokers appears insignificant when compared with the accruing cigarette consumption around the world.

There is ongoing concern regarding the recycling of cigarette butt (CB) waste. The Cigarette Butt Pollution Project in the USA estimated that one-third of the total production of cigarettes was littered to the environment (Cigarette Butt Pollution Project, 2013). While Clean Up Australia (2017) reported CBs as the most common source of rubbish collected in Australia under the miscellaneous category, representing 91.5%. The littering of CBs is a growing waste management issue and requires specific planning. As climate change is becoming the world's biggest challenge, recycling wastes for sustainable development is becoming more crucial (Aldieri et al., 2019).

There has been a great attempt by many Australian companies to develop CB disposal systems. The system involves placing receptacles that come in the form of wall-mounted, post-mounted, and freestanding in areas easily accessible by the public to dispose of their CB waste. Servicing regimes are also provided by companies. The CBs are then collected from the receptacles and disposed of. Landfilling and incineration are the two disposal methods for the collected CBs, however, both methods are costly and pose high exposure risks from hazardous fumes, which can be severely detrimental to the environment and human health (Knox, 2005).

When CBs are littered, harmful chemicals are leached into the environment, posing major concerns for the global environment and the quality of urban and aquatic life. In cigarette smoke, it is possible to classify more than 4000 chemical components that are generated during burning or that are distilled from tobacco, of which 69 are carcinogenic (Hoffmann et al., 2001; Martin, 2018). The main toxic agents include polycyclic aromatic hydrocarbons, formaldehyde, argon, carbon monoxide, nitrogen oxides, benzene, hydrogen cyanide, ammonia, phenol, acetaldehyde, argon, acetone, and pyridines (Hoffmann et al., 2001).

Cigarettes are made of cellulose acetate filters, which are designed to fully or partially retain particulate smoke components, including tar and toxic chemicals (Hoffmann et al., 1995). Cellulose acetate filters have poor biodegradability and can take up to 18 months to decompose under normal environmental conditions, hence, posing a serious dilemma in terms of toxic waste and litter disposal (Ach, 1993; Hon, 1977).

CBs littered on streets commonly get washed away into stormwater drains and end up on beaches, or in rivers and harbors, leaching toxic chemicals (Araújo and Costa, 2019). Marine life regularly mistake CBs for food, and they have been discovered in the stomachs of sea turtles, birds, fish, and other creatures in which they have caused severe digestive problems that have led to death (Rebischung et al., 2018; Slaughter et al., 2011). Marinello et al. (2019), and Torkashvand and Farzadkia (2019) have also reviewed multiple solutions to CB management. Marinello et al. (2019) focused on the advantages and disadvantages of the reviewed CB recycling procedures and suggested proposals for future research. While Torkashvand and Farzadkia (2019) presented suggestions and models for the control of CB littering.

The objective of this paper is to collect and review all major research studies on the toxicity and valorization of CB waste and investigate the feasibility and sustainability of the recycling methods adopted. Particular attention has been given to the toxicity and historical background of CBs, the leachability aspect of the newly derived materials, and the interpretation of engineering properties of two major potential applications (fired clay bricks and asphalt concrete) with graphical results and variations. Eleven CB recycling proposals and practices have been studied, including; (1) mixing in fired clay bricks, (2) mixing in asphalt concrete, (3) production of cellulose pulp, (4) utilization as corrosion inhibitor, (5) preparation of porous carbon, (6) preparation of activated and charred carbon, (7) utilization as sound absorbing material, (8) productions of superhydrophobic sorbent, (9) use as biofilm carrier, (10) preparation of cellulose nano-crystals and (11) use as vector control.

Valorization of CBs in sustainable products may be an environmentally friendly way to manage a continuously growing pollution calamity and a sustainable development for the health of wildlife, humans, marine and freshwater organisms (Aldieri et al., 2019). In some cases, it even implies an enchantment to the modified material. However, based on the review, further research and developments are recommended. In particular, to the methods adopted in handling the CBs and life cycle assessments for all CB recycling proposals.

This systematic review was undertaken by searching papers in ScienceDirect, PubMed, Web of Science, and ProQuest with the aim of finding major investigations performed on recycling CBs. The paper search was carried out in January 2019.

The research was undertaken by identifying relevant keywords specific to cigarette butts (CBs), CB waste, and the application of recycling CBs to obtain novel materials applicable in civil engineering, industrial, agricultural, and chemical or environmental cases. The primary keywords, 'cigarette butt' or 'cigarette' or 'cigarette filter', were searched individually and had to appear in the title or abstract. They were then coupled with the secondary keywords 'recycling' or 'toxicity' or 'littering' or 'health implications' or 'history' or 'composition' to gather quality information and data to prepare a comprehensive review paper. Titles, abstracts, and keywords which included any of the primary, or primary and secondary keywords were studied, and 186 papers were found.

The search process was followed by a selection and screening process to select papers most relevant to the scope of this study. The screening methodology is described as follows:

i. The title and abstract were examined and checked whether the keywords 'CB' or 'cigarette' were consistent. If not, the paper was discarded.
ii. Reliable content that covered interesting technologies on CB recycling was examined and checked whether the methodological process utilized was provided by the authors. Content on the toxicity of cigarettes or toxicity of littered CBs were reviewed and only selected if reliable experiments were conducted. Papers that provided information on the history and composition of cigarettes were thoroughly read and analyzed to make sure data was provided.
iii. Publications that acknowledged the toxicity of CB waste or cigarettes were only selected.
iv. The references in the selected papers were examined and read.

Once all steps were considered and applied, a total of 101 papers were selected and reviewed.

1.1. History of cigarette butts

Tobacco was first consumed in the form of cigars or through chewing tobacco in the late 19th and early 20th centuries. However, during the late 20th century, cigarettes became the preferred tobacco smoking method due to the low-cost and efficiency in nicotine absorption (Wallace et al., 2007). Now, tobacco is one of the most widely used addictive substances in the world, growing in at least 124 countries and occupying more than 4.3 million hectares of land, which is equivalent to the entire land footprint of Switzerland (Eriksen et al., 2015).

By 1950, major concerns began escalating regarding the health effects of smoking. Researchers found a statistical connection between smoking and lung cancer, and a significant number of articles were published in medical and academic journals addressing the negative health effects. Following these findings, the tobacco industry responded by manufacturing low-tar and filtered cigarettes (Hoffmann, 1997). Tar is one of the toxic chemical particles remaining after burning tobacco and constitutes the leading health risks associated with smoking, including cancer, while nicotine is the chemical that causes addiction (Martin, 2018). The filter provides a mouthpiece for smokers and prevents the release of tobacco flakes into the smoker's mouth. Subsequent to the modifications made to cigarettes, the nicotine yield per smoked cigarette decreased from 2.7 to 0.85 mg, and the tar yields decreased from 38 to 12 mg during the period 1953 to 1996. The market share of filtered cigarettes jumped from 5% in 1953 to 20% in 1955, and, by 1975, 90% of the manufactured cigarettes contained filters (Giovino, 2002). However, it should be noted that the reduced yields described here are for machine smoked cigarettes, and not measured in humans.

According to the Washington DC-based Ocean Conservancy's global beach clean-ups, since the development of cigarette filters, CBs have been reported as the most collected waste each year since the beginning of the initiative in 1986 (CNN Health, 2019). A single machine can make up to 20,000 or more filter cigarettes per minute, which is equivalent to 333 cigarettes per second; consequently, 67 cartons of cigarettes are produced every 40 s. In a year if 5.7 trillion cigarettes are manufactured, and about 97% of them contain filters, more than one million tonnes of plastic is being littered into the environment. At present, China is the largest producer and consumer of cigarettes, having consumed 2.35 trillion cigarettes in 2016 (Fig. 1). If current trends continue, cigarette production is expected to increase and, subsequently, CB litter.

2. Composition of CBs

The modern composition of a cigarette consists of a filter, cigarette paper, tobacco, tipping paper, and additives (Fig. 2).

2.1. Paper

The tipping paper, which stabilizes the mouthpiece from saliva, is wrapped around the filter and contains small ventilation holes that allow fresh air to mix with the smoke, thereby diluting the toxic mix of chemicals inhaled (U.S. Food & Drug Administration, 2016); the cigarette paper is below the tipping paper and holds the tobacco blend; as shown Fig. 2. However, once the CB is discarded into the environment, cigarette and tipping paper will still be remaining. The paper may take up to 3–5 months to biodegrade under ambient conditions (Science Learning Hub, 2019).

2.2. Tobacco and additives

The two major subgenera of the botanical genus Nicotiana are N. Rustica and N. Tabacum, which are the main commercial tobaccos (Hoffmann et al., 2001). The N. Tabacum is grown as the primary tobacco crop, which comes as flue-cured (Virginia tobacco), air-cured (burley tobacco), light air-cured, dark air-cured, and

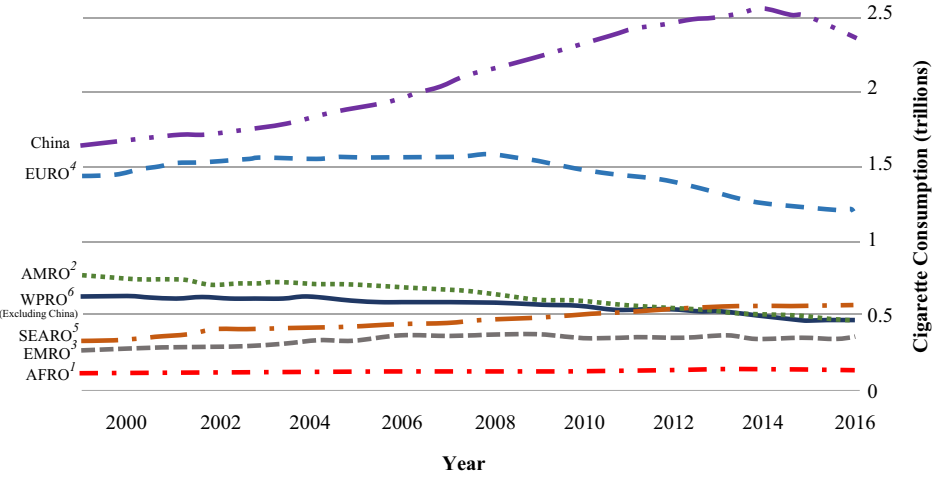


Fig. 1. Global cigarette consumption by WHO region, 1999–2016, in trillions (adapted from The Tobacco Atlas (2018)). [1]Africa. [2]Americas. [3]Eastern Mediterranean. [4]European. [5]Southeast Asia. [6]Western Pacific.

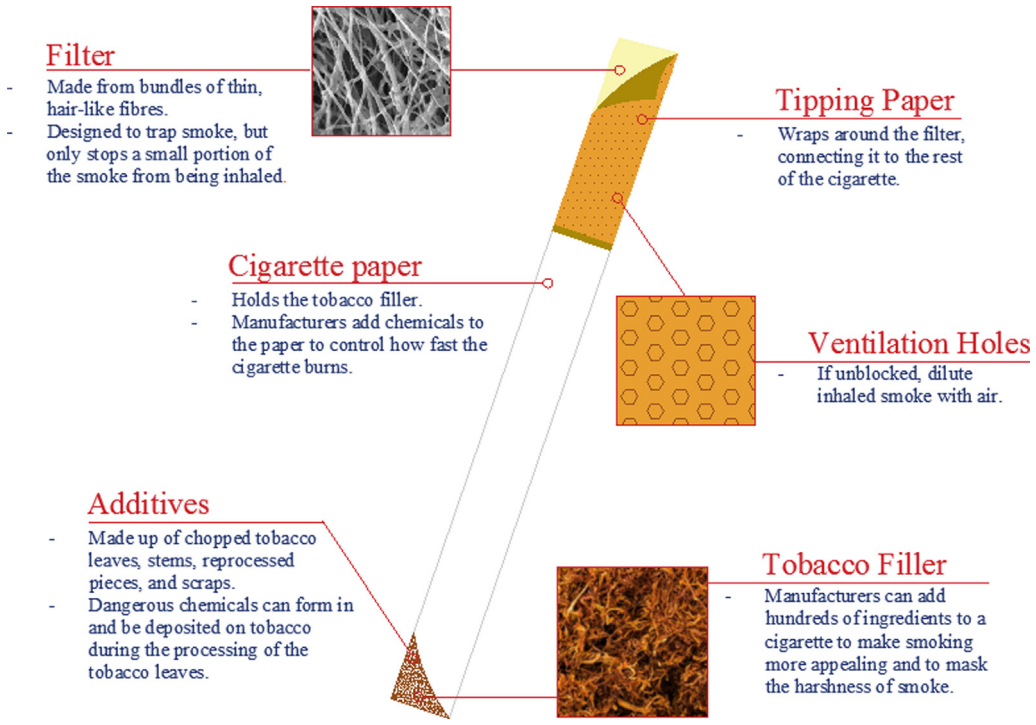


Fig. 2. Physical structure of a cigarette with each component identified and described.

sun-cured tobacco (oriental tobacco) (Tso, 1990). The composition of the tobacco blend varies according to the preferences of the country (Hoffmann et al., 2001). The manufacturing process of the tobacco consist of many stages, including growing the tobacco, harvesting the tobacco, curing the leaf, moistening, stripping, sorting, conditioning, aging, and, finally, blending the leaves (Madehow, 2018). The processing of the leaves produces several by-products, which are processed separately and then added back into the cigarette blend. Additionally, additives are incorporated into the tobacco mixture, including humectants and flavor additives. The primary humectants in cigarette tobacco are glycerol and propylene glycol; their function is to preserve the plasticity and moisture in the cigarette and prevent or delay the drying of tobacco. While common flavor additives include licorice, cocoa solids, and various sugars, these additives act to mask the harsh flavors of smoke (Hoffmann et al., 2001). When a cigarette is discarded into the environment, unburned tobacco residue may remain in the cigarette butt.

2.3. Filter – cellulose acetate

Cellulose is a naturally occurring polymer of glucose ($C_6H_7O_6$) that is derived primarily from wood pulp and cotton linters. Each glucose unit consists of three hydroxyl groups (OH), which can hydrogen bond to adjacent chains; the result is a tough material (Gilbert, 2017). Due to the extended chains of cellulose, it is almost impossible for solvents to penetrate in. The only way to process cellulose is to block the OH groups (hydrogen bonding) with acetate groups. Therefore, by esterifying natural cellulose with acetic acid, cellulose acetate is produced (Fig. 3) (Fischer et al., 2008).

Cellulose acetate is the primary component in the filter of a CB. Therefore the high degree of acetate substitution prevents the cellulose in the littered CB to access the microorganisms (bacterium, fungal, etc.) in the environment for biological decomposition (Petre et al., 1999; Robertson et al., 2012). Ultraviolet (UV) radiation also promotes cellulose acetate degradation, however, the remaining

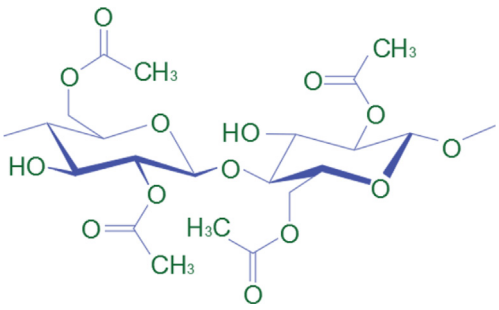


Fig. 3. Molecular structure of cellulose acetate, which is used in the filter of a CB.

tipping paper on the CB inhibits the sunlight UV radiation reaching the cellulose acetate fibers (Robertson et al., 2012). Ultimately the paper biodegrades, and the exposure of the UV radiation to the cellulose acetate surges. However, the composition of the cellulose acetate restricts the exposure of the sunlight to the surface area of the fibers. Additionally, the microorganisms within the environment do not accept the cellulose acetate as a food source (Carollo and Grospietro, 2004). Therefore, de-acetylation proceeds through hydrolysis, however cellulose acetate hydrolysis is a slow course at ambient conditions. Hence why, under anaerobic conditions, it may take up to 1–2 months in soil, 6–9 months in freshwater, and 12 months or more under poor conditions for the CBs to degrade (Ach, 1993; Shenstone, 1995).

3. Toxicity of cigarette butts

When a cigarette is smoked, it is converted to particulate and volatile phases. The particulate phase consists of non-volatile and semi-volatile chemicals, and the vapor phase consists of volatile chemicals. The vapor phase is the portion of the smoke that passes through the filter, and 95% of its weight is made up of 400–500 individual gaseous compounds including nitrogen, oxygen, carbon

dioxide, carbon monoxide, water, argon, hydrogen, acetone, nitrogen oxides, and volatile sulfur compounds (Hoffmann, 1997; Roberts, 1988). The particulate phase is the portion of the smoke that is trapped in the filter and is made up of the remainder of the smoke weight with more than 3500 individual components; these include nicotine, nicotiana alkaloids, compounds specific to solanaceae, catechols, aromatic hydrocarbons, and polycyclic aromatic hydrocarbon (PAH) (Hoffmann et al., 2001; Hoffmann, 1997).

PAH was the first carcinogen identified in a cigarette (Cooper and Lindsey, 1954), and, in 2000, the number of carcinogens identified in tobacco smoking increased to 69 (International Agency for Research on Cancer, 1982). According to the International Agency for Research on Cancer (1982), among the carcinogens, 69 are animal carcinogens, 48 are possibly carcinogenic to humans, 8 are probably carcinogenic to humans, and 11 are proven to be carcinogenic to humans (Hoffmann et al., 2001; Hoffmann, 1997). Hence, cigarette smoking is a serious threat to humans, for every cigarette burned, a smoker receives roughly 1.4–2.2 mg of dangerous compounds (USDHHS, 2010). It is estimated a standard cigarette contains between 9 and 30 mg nicotine, of which 0.5–2 mg is inhaled by the smoker. While, 4–8 mg is the estimated toxic dose of nicotine intake for non-smokers (Malizia et al., 1983). The development of many organ systems can be detrimentally affected by nicotine due to the CBs toxicity, including the respiratory, cardiovascular, nervous, and reproductive systems (Abbott and Winzer-Serhan, 2012; Lushniak et al., 2014).

Smokers and non-smokers may be uninformed regarding the toxicity level of CBs, and the potential hazard the waste may pose to the environment and wildlife (Novotny et al., 2009). When CBs are littered into the environment, significant amounts of toxic chemicals are leached and potentially accumulated in water channels, streets, nature strips, and footpaths, thereby posing a serious threat to local organisms and aquatic species (Novotny et al., 2009). Children are more likely to be exposed to CBs through oral contact as they often like to explore their environments, and 1–2 mg of nicotine is sufficient to cause vomiting and nausea due to the CBs toxicity (Malizia et al., 1983). CBs are reported as the most common toxic substance causing symptoms in children (Novotny et al., 2011).

In 2011, Moerman and Potts (2011) investigated the leaching behavior of heavy metals in CBs. The main purpose of the study was to determine the concentration of metals leached and the effect on the surrounding environment. To understand how the pH level affects leaching behavior, a pH range of 6.00–4.00 was selected, which is typically observed in natural rainwater. It was found the metals studied have different leaching behaviors over time. A direct relationship was found between the concentration of leached metals and the soaking period. An increase in metal concentration leached for Ba, Fe, Mn, and Sr was seen over time, while no significant change in metal concentration was seen for Ti, Pb, Zn, and Ni. In contrast, a decrease in metal concentration for Cu, Al, Cr, and Cd was observed over time. The variable relationships for the selected metals Ba, Zn, and Cu can be seen in Fig. 4. This indicates that a piece of cigarette litter is a point source of metal contamination for a least a month. Moreover, it suggests that the longer the waste remains in the environment, the greater the contamination of Ba, Fe, Mn, and Sr (Moerman and Potts, 2011).

To analyze the toxicity of CBs, the metal concentrations of whole unsmoked cigarettes were measured. The experimental studies exposed more than one-fifth of Mn, Zn, and Ba, one quarter of Cu and half of Sr and Pb may potentially be leached into the environment (Bell and Mulchi, 1990; Iskander et al., 1986). When placed in contrast to the estimated amounts of CBs littered annually (4.95 trillion) (Cigarette Butt Advisory Group, 2009), the results indicate 9.26 ± 0.5 kg (Al), 8.76 ± 2.52 kg (Ba) and 38.1 ± 8.2 kg (Fe) may enter the environment each year from CB litter (Moerman and Potts, 2011).

While Slaughter et al. (2011) conducted a comprehensive study to identify the LC50 (lethal concentration 50) of CB leachate to freshwater and marine ecosystems. The LC50 for leachate from smoked CBs was 0.97 and 1.1 CB/l for both the freshwater fathead minnow and marine topsmelt. While the leachate from smoked cigarette filters exhibited reduced values of 4.3 and 1.8 CBs/l for both fish species. For unsmoked cigarette filters, LC50 values of 13.5 and 5.1 CBs/l were found for fish species (Slaughter et al., 2011). Therefore, the results indicate the toxicity of CB leachate increase from unsmoked to smoked filter.

Chevalier et al. (2018) investigated the possible release of nanoscale particles from CB waste by leaching the waste with rainwater. The results revealed the release of 4.12 ± 0.24% organic carbon in the spectrum of 10 nm up to 400 nm. Moreover, the analysis showed that As, Pb, Cd Cu, Ni, Cr, Co, Al, Mn, Zn, and Fe metallic species were principally attached to the nanoscale particles per gram of carbon released (Chevalier et al., 2018). Hence, the results indicate an evolving global contamination of toxic chemicals, which requires urgent attention and investigation.

To evaluate the genotoxic potential of CBs, Di Giacomo et al. (2016) studied the mutagenicity of CBs in the bacterial reverse

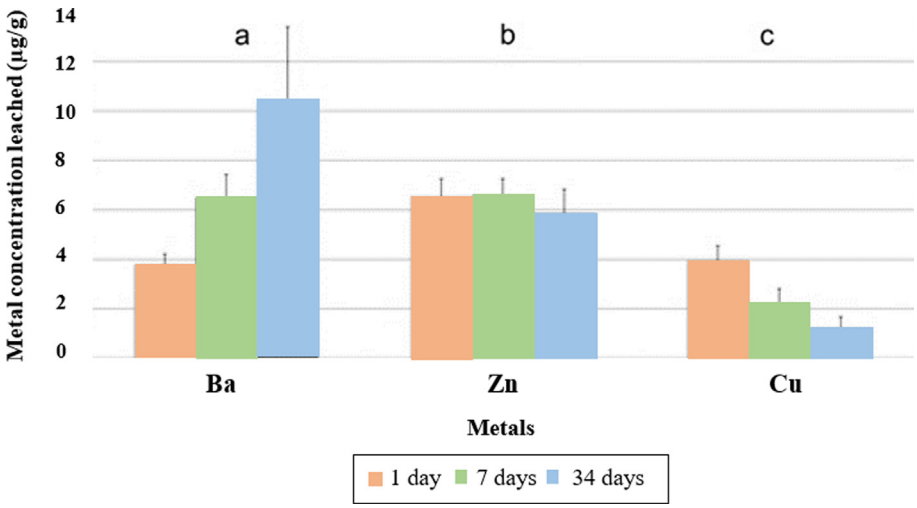


Fig. 4. Relationships observed between metal concentration leached from smoked cigarettes and soaking time in aqueous solution (adapted from Moerman and Potts (2011)).

mutation test on Escherichia coli WP2uvrA and Salmonella typhimurium TA98 and TA100 strains. The experiments were both conducted in the presence and absence of the S9 exogenous metabolic activator. The findings confirmed the mutagenicity of CBs in the presence of the exogenous metabolic activator S9, thereby verifying that CBs contain pro-carcinogenic substances. As a possible preventative strategy, the natural sesquiterpenes β-caryophyllene oxide (CRYO) and β-caryophyllene (CRY) compounds were studied to investigate their ability to inhibit the mutagenicity of CBs. This strategy was based on previous studies proposing the chemopreventative properties of CRYO and CRY (Amiel et al., 2012; Chang et al., 2013; Chavan et al., 2010; Jun et al., 2011; Legault and Pichette, 2007). The experimental results confirmed the antimutagenicity properties of CRY and CRYO on CBs, and, hence, the possibility of using the compound as an environmental decontaminator. Furthermore, the paper suggests utilizing the compound as a maceration solution to apply during disposal for smokers and for use during degradation procedures for CB waste (Di Giacomo et al., 2016).

4. Research on valorisation of cigarette butts

4.1. Mixing in fired clay bricks

Due to the durable nature of the material, bricks have been used in construction for many years. Fired clay bricks are mainly composed of clay and sand, which are bound together under high temperatures in a kiln, making a remarkably strong construction material. Lately, incorporating waste materials in fired clay bricks has been investigated in the hope of offering a sustainable solution to a growing pollution catastrophe. Common waste materials include rubber (Turgut and Yesilata, 2008), polystyrene (Veiseh and Yousefi, 2003), fly ash (Kayali, 2005), sludge (Basegio et al., 2002), sawdust and limestone (Turgut and Algin, 2007), and processed waste tea (Demir, 2008). Moreover, in 2016, Ukwatta et al. (2016) conducted a study based on incorporating biosolids in fired clay bricks. The results indicated that by adding up to 50% biosolids in fired clay bricks, acceptable physical and mechanical properties can be achieved, and up to 25% energy can be saved during the firing process (Ukwatta et al., 2016). Although recycling waste material in fired clay bricks may improve porosity, strength, and shrinkage, further comprehensive testing is required in the field of toxic leaching.

In a recent research, the application of recycling CBs in fired clay bricks were comprehensively investigated (Kadir and Mohajerani, 2012, 2011, 2015; Mohajerani et al., 2016). The experimental procedure involved preparing clay bricks with 2.5%, 5%, 7.5%, and 10% CB content by weight. CBs of varying sizes and brands were provided by Buttout Australia Pty Ltd (Fig. 5). The mixtures were prepared using a Hobart mechanical mixer with mixing times of 5, 10, and 15 min. The samples were then compacted manually in molds using predetermined masses, and the specimens were dried at 105° for 24 h before being fired in a furnace at 1050°.

The samples were tested based on their mechanical and physical properties and compared to control bricks with 0% CB content (Fig. 6). In addition, thermal conductivity, the effects of mixing time, and heating rate on brick properties, energy savings during firing, emission of gaseous compounds, and leachability were tested (Mohajerani et al., 2016).

The property test results show a dramatic decrease in compressive strength from 25.65 MPa with 0% CB content to 3 MPa with 10% CB content (Fig. 7). The dry density decreased from 2118 kg/m^3 for the control sample to 1482 kg/m^3 with 10% CB content, producing lightweight bricks. The water absorption increased from 5% with 0% CB content to 18% with 7.5% CB content. Moreover, the results show a firing energy saving of 30.8% with the incorporation of 2.5% CB content, and a 58.4% energy saving with a 5% CB content (Mohajerani et al., 2016). Additionally, an increase in CB content



Fig. 5. Prepared CBs of varying sizes and brands for recycling (Mohajerani et al., 2016).

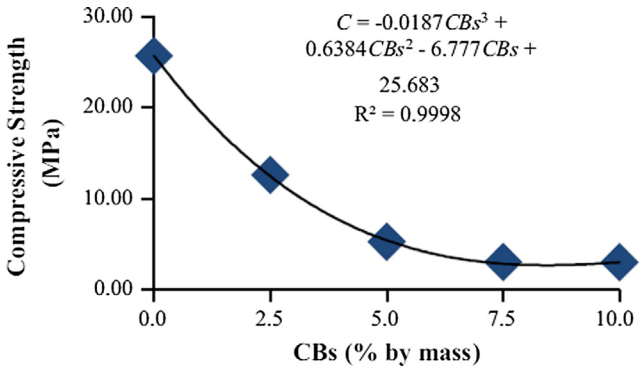


Fig. 7. Compressive strength versus CB% by mass (Mohajerani et al., 2016).

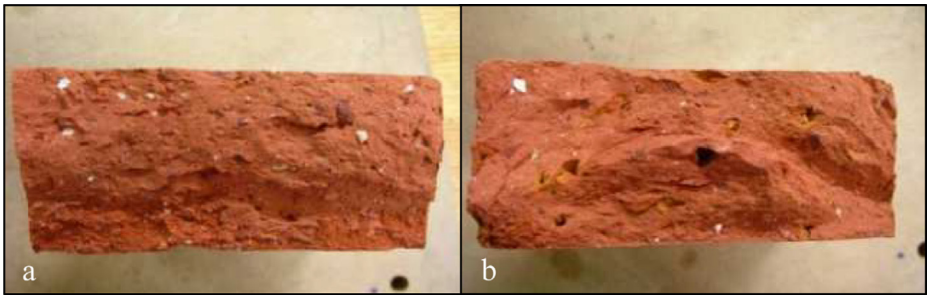


Fig. 6. Cross-sectional view of bricks with 0% (a) and 2.5% (b) CBs (Mohajerani et al., 2016).

resulted in a reduction of thermal conductivity significantly. Therefore, incorporating CBs in the manufacturing of bricks may solve a growing pollution problem, however, further research is essential, including a life cycle assessment.

4.1.1. Leachate test on fired clay bricks incorporated with CBs

The main environmental concern regarding recycling CBs in construction material is the possible leaching of contaminants, such as heavy metals. Therefore, leachate tests are necessary to assess the risk of contaminated leaching. Kadir and Mohajerani (2012) investigated the possible levels of leachability of manufactured fired clay bricks incorporating 0%, 2.5%, 5%, and 10% CB content. Leachate tests were carried out based on the Australian Bottle Leaching Procedure (ABLP), Toxicity Characteristics Leaching Procedure (TCLP), and Static Leachate Test (SLT). The objective of the study was to determine whether the leaching of heavy metals including Arsenic (As), selenium (Se), Mercury (Hg), Barium (Ba), Cadmium (Cd), Chromium (Cr), Lead (Pb), Silver (Ag), Zinc (Zn), Copper (Cu), and Nickel (Ni) from the modified bricks exceeded the regulatory standards. The results exhibited a positive correlation between the leachate of heavy metals and leaching time; by day 134, a decrease in leachate was noted (Kadir and Mohajerani, 2012). Low concentrations in leaching were found, which can be attributed to the high firing temperature during the firing process of the clay bricks incorporating CBs. The tests confirm that all the heavy metal leachate values were insignificant and below the standard acceptable limits.

4.2. Mixing with asphalt concrete

Asphalt concrete is a composite material consisting of coarse and fine aggregates, bituminous binder, and filler (Rebbechi, 2007). The material plays a fundamental role in flexible pavement structures, due to its ability to withstand high volumes of traffic load in various environmental conditions. Due to its composite nature, mixing asphalt concrete with various waste materials has gained widespread attention. The application of waste materials in asphalt concrete mixtures may solve major environmental issues, reduce the demand for natural aggregates, and improve the performance of asphalt concrete (Tahmoorian et al., 2017). Common modes of failure for asphalt concrete include rutting due to excessive deformation, cracking due to fatigue, raveling due to oxidation, and hardening of the binder.

Therefore, methods to improve the properties of asphalt concrete are constantly being investigated. Experiments have confirmed that the incorporation of car tires (Oliver, 1999), glass (Pioneer-Road-Services, 2007), pulverized computer circuit boards (Guo et al., 2008), and nuclear and hazardous wastes (Courtois et al., 1996) have improved the chemical and mechanical properties of asphalt concrete. Paranavithana and Mohajerani (2006) investigated the possible application of recycled concrete aggregates in asphalt concrete. The study found lower resilient modulus, creep, and volumetric property values (except for the percentage of air voids) when compared with the values for fresh aggregate specimens (Paranavithana and Mohajerani, 2006).

In a recent study, a group of researchers investigated the application of CBs in asphalt concrete (Mohajerani et al., 2017). The experiment involved preparing 10 kg/m^3, 15 kg/m^3, and 25 kg/m^3 CBs encapsulated with different bitumen classes (C170, C320, and C600) (Fig. 8) and incorporating them into an asphalt concrete mixture with Class 170 Bitumen (Fig. 9). CBs of varying sizes and brands were provided by Buttout Australia Pty Ltd. The initial step of the experimental procedure involved drying CBs in an oven to eliminate any trapped moisture. Once dried, the CBs were encapsulated with different classes of hot bitumen (C170, C320, and C600) and left to dry (Fig. 8). The aggregates were placed in the oven for



Fig. 8. CBs encapsulated in bitumen (Mohajerani et al., 2017).



Fig. 9. Asphalt concrete samples manufactured with CBs of 10 kg/m^3, 15 kg/m^3, 25 kg/m^3, and no CBs (control) (Mohajerani et al., 2017).

24 h at 105°, once dried, the asphalt mix, the aggregates, and bitumen were placed in the oven and heated for 1 h at 150°. The aggregates were then mixed with the bitumen for 3 min, then the asphalt mix and mold were conditioned for 1 h at 150° in the oven and mixed together with the encapsulated CBs. Once conditioned, the samples were compacted using a gyratory compactor. Following a similar procedure to the first investigation, the second investigation involved assessing the effects of incorporating 10 kg/m^3 of CBs encapsulated with paraffin wax into asphalt concrete mixes manufactured with Classes 170 and 320 bitumen.

Several tests were carried out for both investigations, and the experimental results demonstrated encouraging results. Asphalt mixes with 10 kg/m^3 and 15 kg/m^3 CB content encapsulated with bitumen demonstrated acceptable bulk density, percentage of air voids, and resilient properties. The stability and flow values satisfied the (Rebbechi, 2007) light, medium, and heavy traffic condition requirements of 5.5 kN for light traffic and 6.5 kN for medium and heavy traffic conditions (Fig. 10). Asphalt mixes with 10 kg/m^3 CB content encapsulated with paraffin wax exhibited a decrease in the percentage of air voids and voids in mineral aggregate for C170 samples and an increase for C320 samples (Mohajerani et al., 2017). Both samples satisfied the standard

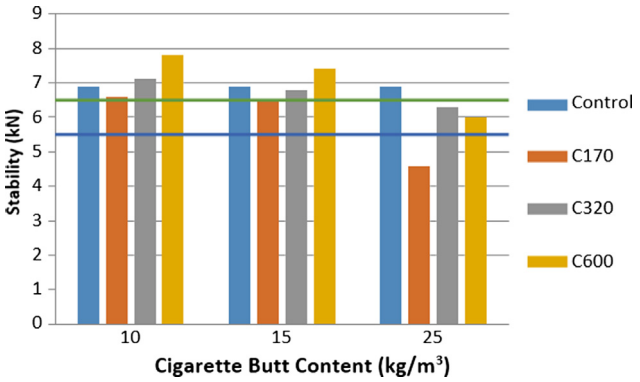


Fig. 10. The effect of different amounts of CBs on the Marshall Stability for CBs encapsulated with different classes of bitumen (C170, C320, C600) (Mohajerani et al., 2017).

requirements of 3–7% for air voids and a minimum of 15% for voids in mineral aggregate, while the stability values only satisfied light traffic conditions for road pavements. In addition, the results suggest a reduction in thermal conductivity, and, subsequently, on the Urban Heat Island effect due to the increase in porosity (Mohajerani et al., 2017). However, further investigations are necessary for the application of CBs encapsulated with paraffin wax and bitumen in asphalt concrete for widespread industry adoption.

4.3. Production of cellulose pulp

Pulp is a lignocellulosic fibrous product manufactured by mechanically or chemically extracting cellulose fibers from renewable wood raw material. A group of researchers in Brazil developed a technology to produce cellulose pulp by recycling CBs (Teixeira et al., 2017). The main idea of the technology is based on separating all the cellulose content in the CB by hydrolyzing the cellulose acetate in the presence of strong alkali (Fig. 11). The experiment involved four different processes, including dissolving all chemicals and ash absorbed in the filter, separating the cellulose acetate into cellulose, eliminating lignin from the residual tobacco and mitigating its cellulose, and, finally, disaggregating the remaining paper (Teixeira et al., 2017).

At the end of the filtering process, cellulose pulp and a dark viscous liquor were produced. The experimental results confirmed that the cellulose pulp could, in fact, be used in the paper industry. However, the dark liquor produced during the pulping process contained very high resistant organic content, including toxic compounds. Several treatment processes were conducted on the effluent, including precipitation by acidification, coagulation, and ozonation to reduce the chemical oxygen demand (COD) value, however, all methods were unsuccessful. Therefore, after precipitating the effluent with a pH of 3, Teixeira et al. (2017) proposed that the clarified effluent should be collected and used in a new pulping process. However, the proposed progression for the clarified effluent fails to offer a sustainable method. Constant production of effluent will be produced, and the effluent produced will be more than what is required for the alkaline pulping, therefore, a practical solution is essential.

4.4. Utilization as corrosion inhibitor

Many investigations have been conducted to determine the corrosion inhibitive efficiency of nicotine (Lucatero et al., 2016), with one study presenting promising results involving applying CBs as corrosion inhibitors for N80 steel at 90° in a hydrochloric acid solution (Zhao et al., 2010a). The investigation involved performing liquid chromatography (LC)/mass spectrometry (MS) and an infrared (IR) reaction analysis system to detect the chemical compositions of CB water extracts to ultimately determine the best aging time. The aging process involved adding five CBs in 100 mL distilled water and continuous monitoring by an LC/MS and an IR reaction analysis system to determine the highest inhibition efficiency of the aging solution. The experimental results recorded 24 h as being the best aging time. To assess the corrosion inhibitive effect of the aged solution on N80 steel at 90° in hydrogen chloride (HCI), electrochemical and weight loss techniques were used. During the investigation, various CB water extracts and HCI concentrations were used to determine their influence on corrosion inhibition. The results presented a maximum corrosion inhibition efficiency of 94.6% in 10% HCI solution with 5% inhibitor. However, it was determined that the increase in HCI solution increased the corrosion attack (Zhao et al., 2010a).

Following this study, Zhao et al. (2010b) compared the corrosion inhibitive effect of anaerobic and aerobic CB water extracts on N80 steel at 90° in HCI to determine the effect of oxidation. Mass spectrometry/liquid chromatography and the infrared reaction analysis system test method were adopted to contrast the chemical compositions of the anaerobic and aerobic extracts. Fewer types of compounds were found in the anaerobic extracts than in the aerobic extracts, thereby indicating that the aerobic condition may favor CBs water extraction over the anaerobic condition. Additionally, to study the corrosion inhibitive mechanism and assess the corrosion inhibitive effect of both anaerobic and aerobic CB water extracts, potentiodynamic polarization, X-ray diffraction, weight loss, electrochemical noise, and impedance were used. The results revealed that the aerobic extracts consisted of more compounds and had greater corrosion inhibitive effects than the anaerobic extracts (Zhao et al., 2010b).

To further enhance the inhibition efficiency, Zhang et al. (2011) suggested incorporating cuprous chloride (CuCI) to inhibit the cor-

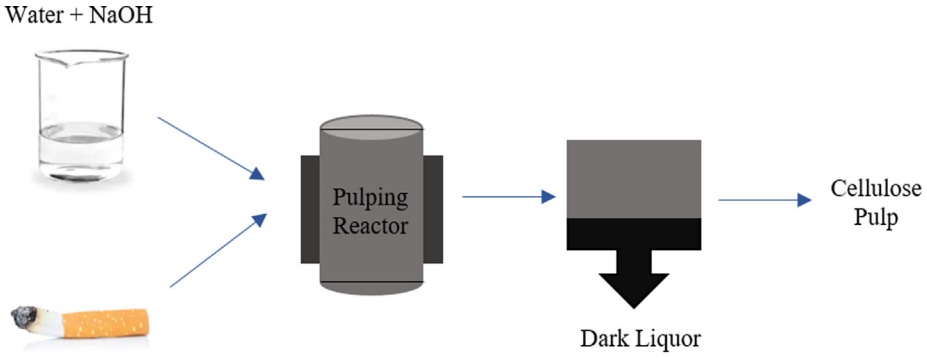


Fig. 11. Cellulose pulp production flow chart from CBs.

rosion of N80 steel in HCl solution. The study involved comparing the corrosion inhibition of water extracts from CBs in the presence and absence of CuCl on N80 steel at 90° in 15% HCl. In addition to the tests performed in the previous two papers, an X-ray photoelectron spectroscopy test was conducted to examine the composition of the protective coat formed by the CuCl. The results revealed an increase in inhibition efficiency of 95.3% with the addition of CuCl and 9% CB water extract (Zhang et al., 2011). In addition, the X-ray photoelectron spectroscopy results confirm that the copper added water extract from CBs improved the N80 steel surface with a protective coat. Although the results seem promising, the paper requires further research. The experimental procedure requires further clarification, and a comprehensive study is required on a leachate analysis as the paper fails to mention the prospect of toxic leaching during the lifetime of the modified material. A summary of all research conducted on the application of CBs as a corrosion inhibitor on N80 steel can be seen in Table 1.

4.5. Preparation of porous carbon

Porous carbon is a versatile material and can be used for several applications due to the material's high porosity, relative easy surface chemical tuning, large surface area, and excellent electron conductivity (Matos et al., 2017). However, the preparation involves expensive and/or toxic chemicals and sophisticated techniques. Polarz et al. (2002) found a simple one-step procedure for preparing porous carbons, which involves inexpensive precursors, cellulose acetate in the form of CBs. In conducting extensive experiments, it was found that complex biomolecules like cellulose acetate can produce porous carbons that exhibit well-defined positions on different length scales ranging from the nanometre to the micrometre (Polarz et al., 2002). Based on these findings, Lee et al. (2014) and Yu et al. (2018) investigated recycling used CBs as a proper carbon source for use in super-capacitors. The process involved heat treating used CBs under nitrogen (N) containing atmosphere to prepare N-doped mesoporous and microporous hybrid carbon material (NCF) through a one-step method; pyrolysis (carbonization) (Fig. 12). The combination of an enlarged surface area, the pseudocapacitive behavior, and well-developed mesopores and micropores resulted in achieving the desired supercapacitive performance (Lee et al., 2014).

Similarly, Ghosha et al. (2017), and Yazdi et al. (2012) prepared a conducting material from used CBs through heat treatment; the study required no pre-treatment for the CBs prior to the carbonization process. The process involved characterizing and investigating the electrochemical properties of the pyrolyzed material through dynamic light scattering, Raman spectroscopic, X-ray diffraction, ultra-violet visible spectroscopy, Fourier transform infrared spectroscopy, zeta potential measurements, and field emission scanning electron microscopy (Yazdi et al., 2012). The results confirmed the prospect of using the pyrolyzed material in electronic applications as the material demonstrated good conductivity (Ghosha et al., 2017). However, further investigations are required in this field of study for widespread industry adoption.

4.6. Preparation of activated and charred carbon

Hamzah and Umar (2017) investigated the potential of preparing active carbon from CB waste through microwave induced KOH activation. The microwave heating method was adopted in preparing active carbons due to the method's rapid and uniform heating capacity. The process involved an activation time of 20 min at 630 W and a frequency of 2.45 GHz. The paper proposed that the resultant active carbon can be used as an alternative absorbent material for oil-water separation, supercapacitors, and methylene blue due to its structural diversity and large surface area (Hamzah and Umar, 2017).

Table 1
Applying CBs as a Corrosion Inhibitor for N80 Steel.

Preparation of Inhibitors from CBs	CBs	HCl	Modifier	Corrosion Inhibitive Evaluation Techniques	Corrosion Inhibition Efficiency	Reference
5 CBs are added into 100 mL distilled water and aged for 24 h	5%	10%	–	• Weight loss • Potentiodynamic polarization • Impedance measurement	94.6%	(Zhao et al., 2010a)
5 CBs are added into 100 mL of distilled water and aged under bubbling air conditions for 24 h	5%	15%	–	• Weight loss • Electrochemical noise • Potentiodynamic polarization • Impedence • X-ray diffraction	92.9%	(Zhao et al., 2010b)
5 CBs are added into 100 mL of distilled water and aged for 24 h and reacted with 0.5 g of CuCl at 60° for 2 h.	5%	15%	9% CuCl	• Weight loss • Electrochemical noise • Potentiodynamic polarization • Impedence • X-ray photoelecetron spectroscopy	95.3%	(Zhang et al., 2011)

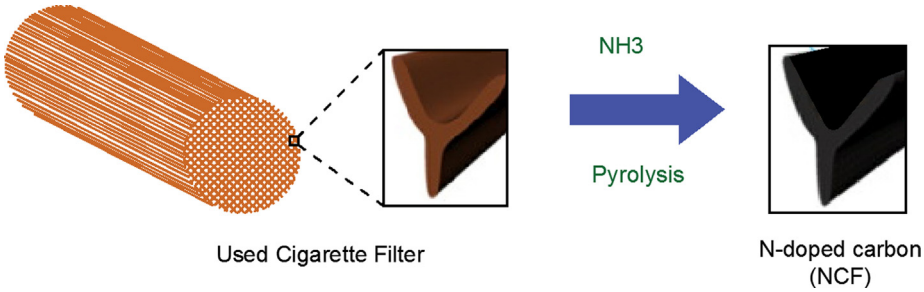


Fig. 12. Fabrication procedure used for preparing NCF by pyrolysis of a used cigarette filter.

Masoudi Soltani et al. (2015) investigated producing charred carbon through the pyrolysis of CBs. The charred carbon was modified with NHO_3, and a decrease in surface area and micropores was observed. The study analyzed the efficiency of using non-modified and nitric acid-modified charred carbon to remove lead from the aqueous solution (Masoudi Soltani et al., 2015).

4.7. Utilization as sound absorbing material

In a recent research, Escobar and Maderuelo-Sanz (2017) proposed using CBs as an absorbing material due to the porous nature of the waste (Matos et al., 2017). The investigation involved analyzing the acoustical performance of a new kind of material using used and unused CBs. In addition to studying the effects of compression, the influence of the length of butts, and a comparison with other materials. The results revealed a high absorption coefficient of <0.90 from mid-frequencies (>2000 Hz), indicating that CBs can be used as a substitute for artificial based commercial products (Escobar and Maderuelo-Sanz, 2017). The samples used for this study comprised intact CBs, however, CBs contain toxic chemicals, and, therefore, require pre-treatment. Maderuelo-Sanz et al. (2018) investigated the acoustical performance of treated CBs. The CBs were initially dried at 80° and sealed in plastic bags; this process was conducted to disinfect the waste from any toxic chemicals and remove humidity. The physical properties, including fiber diameter, porosity, skeletal density, and flow resistivity, were measured. The acoustical performance of the new material was assessed and compared to other fibre materials, the results revealed that the performance of both materials were rather similar, however, the CB material demonstrated better performance (Maderuelo-Sanz et al., 2018).

4.8. Production of superhydrophobic sorbent

Ou et al. (2016) found that by hydrolyzing cellulose acetate fibers from CBs in a sodium hydroxide (NaOH) solution, and, subsequently, in an ethanolic solution of hexadecyltrimethoxysilane, superhydrophobic/ superoleophilic fibers (SFs) can be produced. The produced SFs can be used for oil spill clean-up due to its wetting behaviors towards kerosene and water. The study involved testing the clean-up efficiency of the SFs in a staged kerosene spill, and it was found that the SFs had a 96% efficiency in absorbing the kerosene. Additionally, the moisture content in the collected SFs was found to be lower than the kerosene poured onto the water. Hence, the SFs may be used in the clean-up of oil spills, and, simultaneously, overcome the detrimental effects of CBs (Ou et al., 2016).

Similarly, Ifelebuegu et al. (2018) and Xiong et al. (2018) investigated a two-step process to convert waste cigarette filters into a oleophilic and superhydrophobic sorbent for the application in spill clean-up and water/oil separation. The process involved cleaning the filters ultrasonically, followed by modifying the surface through chemical vapor deposition using methyltrichlorosilane. Ifelebuegu et al. (2018) found the functionalized waste filters can exhibit superhydrophobic properties, absorbing 16 to 26 times their weights in numerous oils with up to 20 cycles of reuse capacity. While Xiong et al. (2018) discovered an absorption capacity of 80% and 82% for pump and silicone oil after 10 cycles.

A similar investigation on recycling cigarette filters for oil-water separation was conducted by Liu et al. (2015, 2019), which involved preparing cigarette filter coated meshes (CFCMs) through a facile electrospinning approach. The CFCMs achieved a performance of underwater superoleophobicity without the need for any chemical modification. An extraordinary separation efficiency larger than 99.9% for water/oil mixtures was achieved after many cycles (Liu et al., 2015, 2019).

4.9. Use as biofilm carrier

A study was carried out to evaluate waste cigarette filters in tobacco factories as a biofilm carrier in the integrated fixed film activated sludge (IFAS) process (Sabzali et al., 2011). Biofilm reactors are used for the removal of nitrogen species and organic carbon in biological wastewater treatment. Hence, this study aimed to develop a new, efficient, and inexpensive carrier with the use of cigarette filters. The investigation confirmed the effectiveness and feasibility of using cigarette filter rods as a biofilm carrier in an IFAS system to simultaneously remove organic matters and nutrients from wastewater (Sabzali et al., 2011). Following this study, Sabzali et al. (2012) investigated the use of cigarette filter rods as a biofilm carrier in an anaerobic moving bed biofilm reactor (AMBBR). The study revealed that cigarette filter rods are feasible and effective as Kaldnes media for the removal of phosphate and organic matter from wastewater. Additionally, the new bi-carrier is cost effective and durable and can increase the effective mixed liquor suspended solids in an anaerobic reaction basin due to the modified materials roughness, high surface area, and high porosity properties (Sabzali et al., 2012).

4.10. Preparation of cellulose nano-crystals

In a recent research, Abu-Danso et al. (2019) investigated the use of CBs for the removal of diclofenac from water. To obtain phosphate ions (HPO) – cellulose nano-fibers (CNFs), acetyl groups were cleaved through alkali treatment and etched with HPO from phosphoric acid. While sulfuric acid was used under high temperatures to cleave phenol moieties from CNFs to obtain cellulose nano-crystals (CNCs). The experiments revealed a maximum removal capacity of 107.9 mg g^{-1} of Dfc through the synthesis of HPO-CNFs, suggesting that celluloses extracted from CBs can potentially be used as an absorbent for the removal of Dfc from water (Abu-Danso et al., 2019).

4.11. Use as vector control

A number of strategies have been developed to control vectors in dengue epidemic areas with insecticides (World Health Organization, 2012). However, the species resistance has resulted in little success. CBs abound in chemical products, which are known to have pesticidal, fungicidal, insecticidal, and herbicidal properties (Murugan et al., 2018). Therefore, in search of an alternative control method, Dieng et al. (2011) and Mondal et al. (2015) found CB waste to be a potential source to manage mosquito-borne diseases. The study involved examining the developmental alterations of Aedes albopictus in the presence of CB waste (Dieng et al., 2011). The results exhibited an increase in the mortality of Aedes albopictus when subjected to CB waste solutions. Subsequently, Dieng et al. (2013) assessed the vulnerability of Aedes aegypti to CBs during their development and found an interruption of larval development when subjected to CB waste. They tested modifications in egg hatching, longevity, reproductive capacity, fecundity, and ovipositional behaviors. Based on this study, Dieng et al. (2014) further investigated the effects of CB exposure on the dengue vector Aedes aegypti to examine whether the waste may alter any biological parameters of parents and their progeny. The results indicate that CB waste can be developed as a novel vector control strategy as it has an indirect effect on adult life traits (Dieng et al., 2014). However, there are many unknown factors within this area of study, and, therefore, further research is essen-

Table 2
Summary of some of the significant research conducted on the valorization of CBs.

CB Recycling Methods	Processing Used	Remarks	References
Mixing in fired clay bricks	No Processing	• The addition of CBs demonstrated a decrease in dry density producing lightweight bricks. The CBs completely burnt off during the firing process. • A firing energy saving of 30.8% with the incorporation of 2.5% CB content was found due to the high heating value of CBs. • An increase in CB content resulted in a reduction of thermal conductivity significantly due to the increase in porosity. • The paper proposes that the incorporation of 1% CBs in the manufacturing process of fired clay bricks can completely offset the CBs produced in the world. • Further research is recommended, including a life cycle assessment.	(Kadir and Mohajerani, 2012, 2011, 2015; Mohajerani et al., 2016)
Mixing in asphalt concrete	No Processing	• Asphalt mixes with CB encapsulated bitumen, and paraffin wax demonstrated acceptable bulk density, stability, percentage of air voids, and resilient properties. • Results suggested a reduction in thermal conductivity and subsequently on the Urban Heat Island Effect. • Substituting CB waste in the production of asphalt concrete will result in a decrease in demand for natural aggregates. • A leachate analysis and life cycle assessment are essential.	(Mohajerani et al., 2017)
Production od cellulose pulp	Hydrolyzing the CBs for 3 hrs in the presence of NaOH	• The extracted cellulose pulp from the CBs can be used instead of renewable wood raw material in the paper industry. • A dark liquor is produced containing very high resistant organic content, including toxic compounds. • A solution is required for the disposal or reuse of the dark liquor.	(Teixeira et al., 2017)
Utilization as corrosion inhibitor	CBs aged for 24 h	• A corrosion inhibition efficiency of 94.6% and 92.9% for 10% and 15% HCI • A corrosion inhibition efficiency of 95.3% for 15% HCI and 9% CuCI • The corrosion inhibitor can be used on N80 steel in construction. • Leachate analysis is required for the CB extracted metal surface.	(Zhang et al., 2011; Zhao et al., 2010a, 2010b)
Preparation of porous carbon	Hydrolyzing the CBs in nitrogen under heat	• Porous carbons that exhibit well-defined positions on different length scales ranging from the nanometre to the micrometer was produced. • The CBs can be used as a proper carbon source in supercapacitors and in electronic applications as it demonstrated good conductivity. • The experimental method involves the removal of the coated paper to obtain the filter. However, on a large scale, this method is not practical. Moreover, the disposal method of the coated paper is not discussed. • Another experimental method consists of utilizing only half of the filter to exclude the effect of pre-contained activated carbon, however the treatment/disposal of the other half is not considered.	(Ghosha et al., 2017; Lee et al., 2014; Polarz et al., 2002; Yazdi et al., 2012; Yu et al., 2018)
Preparation of activated and charred carbon	Microwaving the CBs Pyrolysis of CBs in NHO_3	• Active carbon can be used as an alternative absorbent for oil-water separation, supercapacitors, and methylene blue due to its structural diversity and large surface area. • Charred carbon can be used to remove lead from aqueous solution • The research proposes to wash CBs in hot water and unwrap from the cigarette paper, on a larger scale, this method is impractical to achieve.	(Hamzah and Umar, 2017; Masoudi Soltani et al., 2015)
Utilization as sound absorbing material	Manual removal of cellulose acetate from the CB	• The product can be used as an absorbing material due to the modified materials elevated acoustical performance. • The results revealed a high absorption coefficient of < 0.90 from mid-frequencies (greater than2000 Hz). • The filter can be used as a substitute for artificial based commercial products. • The toxicity of the filter is not considered, therefore further research is essential to investigate the release of heavy metals and toxic chemicals during the use of this new material.	(Escobar and Maderuelo-Sanz, 2017; Maderuelo-Sanz et al., 2018)
Production of superhydrophobic sorbent	Hydrolyzing the CBs in NaOH and ethanolic solution of hexadecyltrimethoxysilane. Chemical vapor deposition using methyltrichlorosilane on CBs. Facile electrospinning on the CBs.	• The produced superhydrophobic fibers can be used for the clean-up of water/oil spills • The processed cellulose acetate fibers demonstrated acceptable wetting behaviors towards kerosene and water. • The filters absorbed oil 16 to 26 times their weight. • A 99.9% water/oil mixture separation efficiency was achieved after many cycles.	(Ifelebuegu et al., 2018; Liu et al., 2015, 2019; Ou et al., 2016; Xiong et al., 2018)

Table 2 (*continued*)

CB Recycling Methods	Processing Used	Remarks	References
		• A life cycle assessment is necessary to assess the chemical characteristics of the superhydrophobic fibers once it deteriorates its sorption capacity and needs to be disposed of.	
Use as biofilm carrier	Integrated fixed-film activated sludge process used on the CBs	• The cigarette filter rods can be used as effective Kaldnes media for the removal of phosphate and organic matter from wastewater. • Further research is essential including a leachate analysis and life cycle assessment.	(Sabzali et al., 2011, 2012)
Preparation of cellulose nano-crystals	Acetyl groups of CBs cleaved through alkali treatment and etched with HPO from phosphoric acid	• The celluloses extracted from CBs can potentially be used as an absorbent for the removal of diclofenac from water. • A maximum removal capacity of 107.9 mg g^{-1} of Dfc through the synthesis of HPO-CNFs was achieved. • Further research is essential including a leachate analysis and life cycle assessment.	(Abu-Danso et al., 2019)
Use as vector control	Soaking of CBs in distilled water	• An increase in mortality of Aedes albopictus was found in the presence of CBs. • CB waste can be developed as a novel vector control strategy as it has an indirect effect on adult life traits of Aedes aegypti. • To specifically target dengue vectors in an open environment is a complex process that requires extensive research.	(Dieng et al., 2013, 2014, 2011; Mondal et al., 2015)

tial. The valorization of CBs into these 11 novel materials has been investigated and summarized in Table 2.

5. Discussion and conclusion

The estimated volume of produced cigarettes worldwide, the persistency, as well as the toxic content after usage, support the necessity for efficient disposal and recycling strategies to solve a global CB waste catastrophe. Existing disposal strategies for collected CB waste include incineration and landfilling. However, both techniques are considered unsustainable and inefficient due to environmental and economic concerns. Therefore, this study has reviewed possible technology on the valorization of CB waste and examined the viability of the research conducted. Eleven CB valorization proposals and practices have been studied, including; (1) mixing in fired clay bricks, (2) mixing in asphalt concrete, (3) production of cellulose pulp, (4) utilization as corrosion inhibitor, (5) preparation of porous carbon, (6) preparation of activated and charred carbon, (7) utilization as sound absorbing material, (8) productions of superhydrophobic sorbent, (9) use as biofilm carrier, (10) preparation of cellulose nano-crystals and (11) use as vector control.

Based on the extensive research conducted on the toxicity of CBs, it is clear the chemical and physical characteristics of CBs can potentially be utilized. However, this does not endorse the practicability and sustainability of the recycling method adopted. Hence why it is crucial to conduct a life cycle assessment of any new material recycled with CBs to prevent the beginning of a new waste catastrophe for the environment.

Throughout the study, various CB processing methods were reviewed, and a vast majority of the recycling methods made use of the chemicals trapped in the CB, or the cellulose acetate filter. In contrast, to a minority of studies utilising the CB as a whole, without employing a processing technique. In summary, the following conclusions have been derived from this investigation:

- A practical recycling method involved incorporating CBs in the production of bricks. Whereby, the literature reviewed proposes that all CBs can sustainably be recycled in fired clay bricks by incorporating 1% CBs in 2.5% of the world's brick production. Additionally, the experimental results were found to be within the standard limits for various properties, including strength, water absorption, efflorescence, and shrinkage, which are important parameters when manufacturing bricks. Moreover, the leaching of contaminants from CBs in bricks was found to be negligible. However, further research is recommended, predominantly a life cycle assessment.
- CBs have the potential to be used as an aggregate in asphalt concrete. Asphalt concrete is a combination of aggregates and asphalt cement, whereby the aggregate constitutes 90 – 95% of the total mixture. Therefore the replacement of CBs in the production of asphalt concrete for natural aggregates can offer a sustainable alternative to reduce the demand for natural aggregates. Due to the toxic nature of CBs, the encapsulation process will act to prevent the leaching of those chemicals by inhibiting water from reaching the filters. Moreover, the physical and mechanical properties generally demonstrated acceptable bulk density, percentage of air voids, and resilient properties, while the stability values only satisfied the light traffic conditions for road pavements. However, research on CBs in asphalt concrete is limited, therefore further experimental work, including a leachate analysis and life cycle assessment, is essential.
- Extracting the cellulose fibers from the cellulose acetate for use in the pulp and paper industry offers a sustainable solution to deforestation. However, the process consequently leads to the production of wastewater containing toxic chemicals from the cellulose acetate. Despite the inconclusive nature of the results, the extracted cellulose pulp from the CBs can decrease the harvesting of trees for paper manufacturing. Moreover, the porous nature of CBs allows it to be a perfect absorbing material with exceptional acoustical performance. Which can be used as a substitute for artificial based commercial products. However, the literature reviewed requires clarification on the pretreatment process of the CBs due to the highly toxic nature of the waste.
- The cellulose acetate in the CB is an important target for CB recycling. A vast majority of researches has utilized the porous nature of the waste for water and wastewater treatment applications. Including the fabrication of superhyrophobic fibers for clean-up of water/oil spills; biofilm carrier for the removal of phosphate and organic matter from wastewater; cellulose nanocrystals and nanofibres for the removal of diclofenac from water; active carbon for oil-water separation, and charred carbon for the removal of lead from aqueous solution. Although

there is various equipment available for wastewater treatments, the utilization of CBs offers an innovative and sustainable approach. It is important all components of the CB is considered and not discarded as waste or require further management.

- Porous carbon derived from CBs are of high surface area with excellent electron conductivity and can be used for supercapacitors and electronic applications. Porous carbon can be used in several applications, however, is considered expensive and consists of sophisticated techniques. Therefore recycling CBs to produce porous carbon is a practical technique, however, the research conducted requires further experimentation and clarification. The methods adopted fail to recycle the whole CB, but rather attempt to make use of only one component of the waste and discard the remaining.
- The nicotine content trapped in the CB can be extracted to be used as an effective corrosion inhibitor for metals. However, applying the extract on metal raises environmental concerns for the possible leachate of toxic chemicals. Similarly, utilizing CBs grounded on its pesticidal, fungicidal, insecticidal, and herbicidal properties for vector control may result in the release of pollutants during the recycling process.

In conclusion, the valorization of CBs may be an environmentally friendly way to manage a continuously growing pollution calamity. In some cases, it even implies an enhancement to the modified material. The aim of this review was to systematically evaluate the toxic nature of CBs, comprehensively review all major research studies on recycling CB waste, and investigate the feasibility and sustainability of the recycling methods adopted. Based on the review, the following research and developments are recommended:

- A life cycle assessment to evaluate the positive and negative environmental impacts of the proposed CB recycling procedures from the collection of the waste through to the production process and finally, the packaging and storage.
- The preparation and implementation of new policies for different recycling options, through the affiliation of government agencies, the environmental protection authority (EPA), and waste recycling industries. In relation to the collection, safety, and storage of CBs; the processing procedures of the various applications; toxicity and leachability; long term conditions and disposal.
- The evaluation of employee exposure, transport, collection processes, and consumer exposures for all recycling schemes mentioned.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work is part of an ongoing postgraduate study on recycling cigarette butts in fired-clay bricks. The authors would like to thank Butt-Out Australia Pty Ltd and the School of Engineering, RMIT University, for their financial and in-kind support.

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